Fabric Workspace Architecture & Automated Pipeline Implementation

Research question:

Fabric workspace architecture

Proven frameworks to organize fabric workspaces for different environments, dev test, prod).

1. Approach

The key focus was on exploring popular workspace architecture models:

1. Medallion Model – Best suited for analytics-focused teams.

2. Domain Aligned – Enterprise layers.

2. Hybrid Model (Medallion + Enterprise layers) – Ideal for enterprise-level deployments.

For the implementation, Hybrid Model is taken.

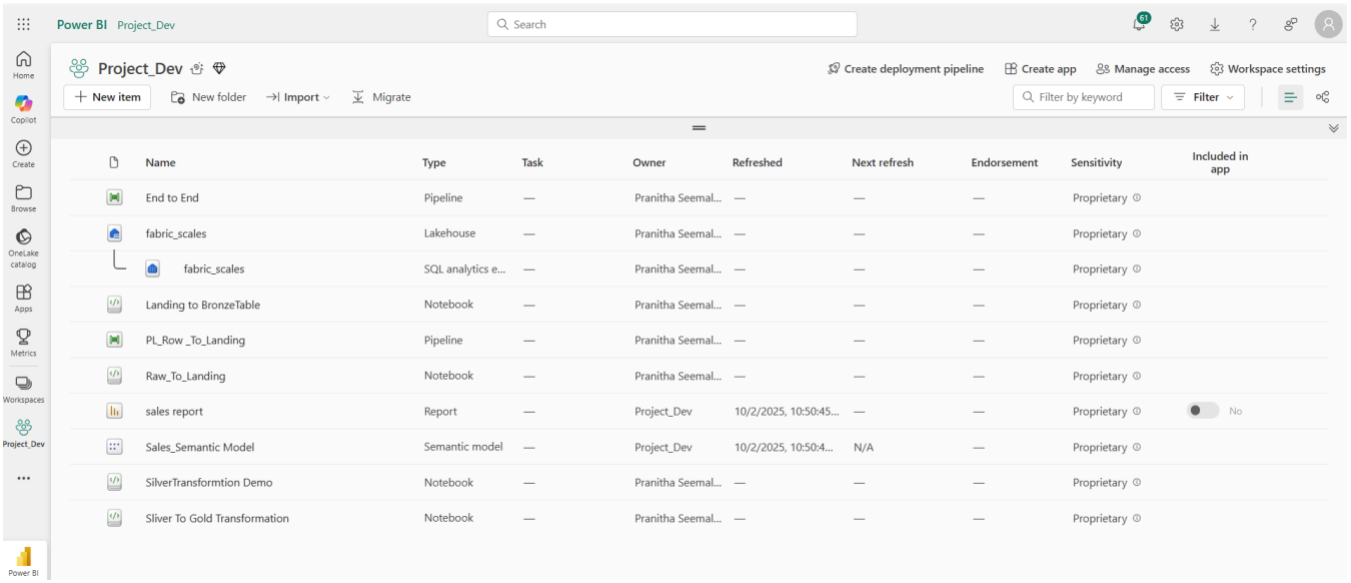
2. Workspace Architecture

Development Environment: Created a workspace Project_Dev for development and testing.

- Data Lakehouse: Ingested and organized sales dataset(listed

the sources below) into a Lakehouse structure (bronze, silver, gold layers).
- Pipelines: Designed automated pipelines to orchestrate data flow across layers [end to end].
- SQL Analytics Endpoint: Enabled for querying and analysis.
- Notebooks: 1. Raw to landing.
2. landing at the bronze table.
3. bronze to sliver.
4. sliver to gold.

-Piplelines: 1.Pl_Row_To_Landing
2. End to End.



Workspace Design Approach:
- Dev-Test-Prod separation ensures scalability, version control, and collaboration.
- Medallion Layering (Bronze → Silver → Gold) ensures clean,

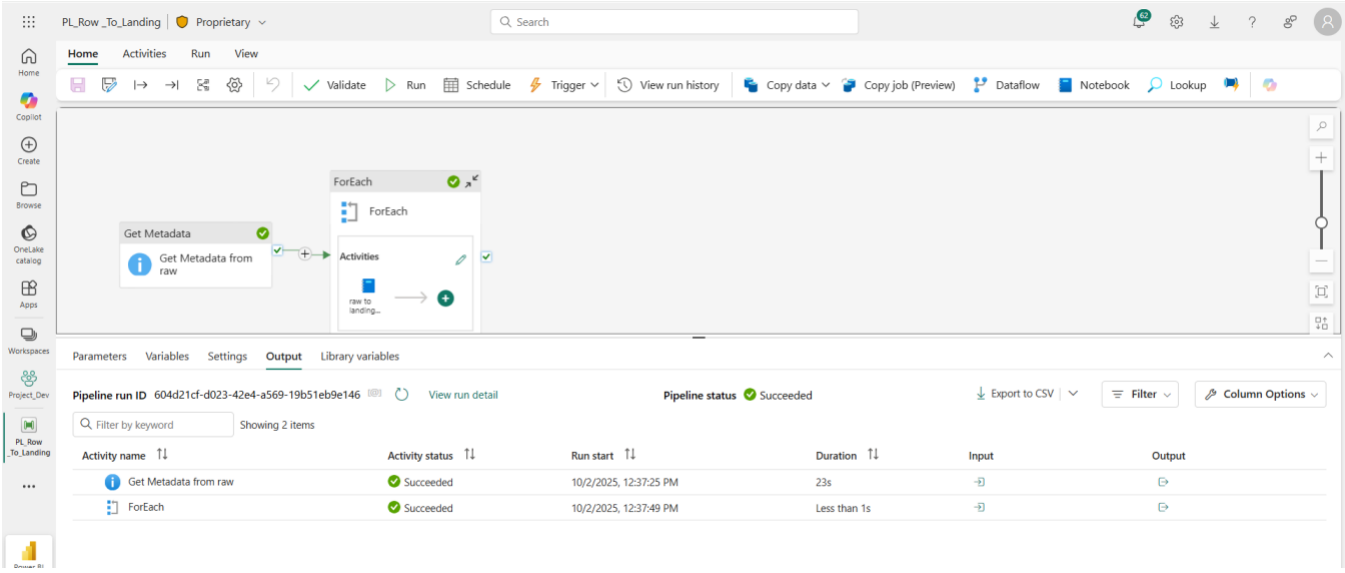
validated, and business-ready data.
- Hybrid Consideration allows extending to enterprise data governance needs.

3. Work Performed

Step 1: Workspace & Environment Setup
- Created Project_Dev workspace in Microsoft Fabric.
- Configured environment for development with scope to extend for other environments.

Step 2: Lakehouse & Dataset Integration
- Integrated fabric_scales dataset into Fabric Lakehouse.
- Built structured tables: dim_customer, dim_product, fact_sales, tdlsales bronze and tblsales sliver.
- Implemented bronze (raw data), silver (cleaned data), and gold (business-ready data) layers.

Step 3: Pipeline Development
Designed an End-to-End pipeline with the following stages:
1. Raw to Landing (Ingestion of source data)
2. Landing to Bronze (Storing raw structured data)
3. Silver Transformation (Cleaning, joining, standardizing datasets)
4. Silver to Gold (Final transformations for reporting and insights)



Step 4: SQL & Analytics Enablement

- Exposed a SQL Analytics Endpoint for business queries.
- Enabled direct integration with BI tools for reporting.

4. Key Findings

- Medallion architecture works efficiently for analytics teams, allowing structured incremental data refinement.
- Hybrid + Medallion is a scalable approach for enterprise adoption, balancing governance with agility.
- The pipeline successfully automated end-to-end data processing, reducing manual interventions.
- Queries on gold layer datasets showed optimized performance, suitable for business dashboards.

5. Business Impact

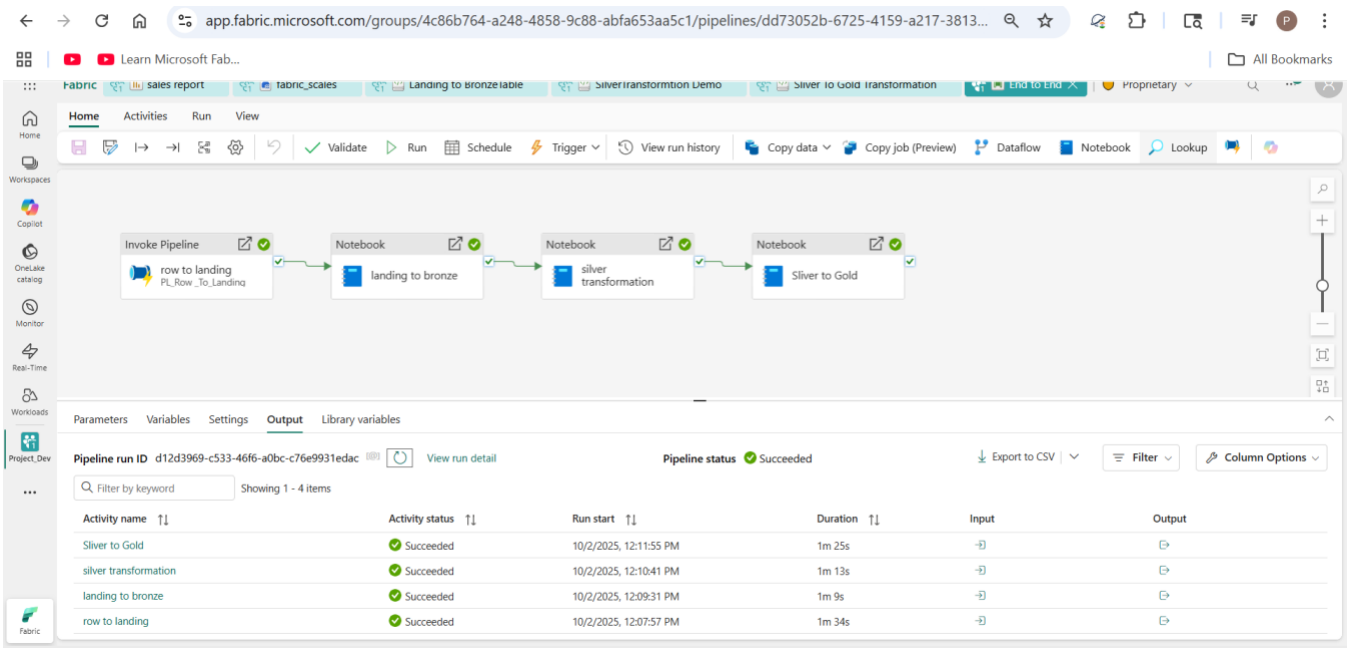
- Improved Data Quality: Standardized data transformation through Bronze → Silver → Gold pipeline.

- Automation: Reduced manual overhead by implementing scheduled pipelines.
- Scalability: Architecture supports both team-level analytics (Medallion) and enterprise adoption (Hybrid).
- Faster Insights: Business-ready gold datasets accelerate dashboard creation and reporting.
- Governance & Security: Workspace design supports separation of environments with sensitivity labels applied.

6. Next Steps / Recommendations

Extend current development workspace into Test and Production environments.
- Implement data refresh schedules to automate pipeline runs.
- Integrate with Power BI dashboards for business reporting.
- Explore materialized views for performance optimization.

Working model:


app.fabric.microsoft.com/groups/4c86b764-a248-4858-9c88-abfa653aa5c1/pipelines/dd73052b-6725-4159-a217-3813...
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Run
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Trigger
View run history
Copy data
Copy job (Preview)
Dataflow
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Lookup
Invoke Pipeline
row to landing
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Notebook
landing to bronze
Notebook
silver transformation
Notebook
Sliver to Gold
Parameters
Variables
Settings
Output
Library variables
Pipeline run ID d12d3969-c533-46f6-a0bc-c76e9931edac
View run detail
Pipeline status Succeeded
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Activity name
Activity status
Run start
Duration
Input
Output
Sliver to Gold
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silver transformation
Succeeded
10/2/2025, 12:10:41 PM
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row to landing
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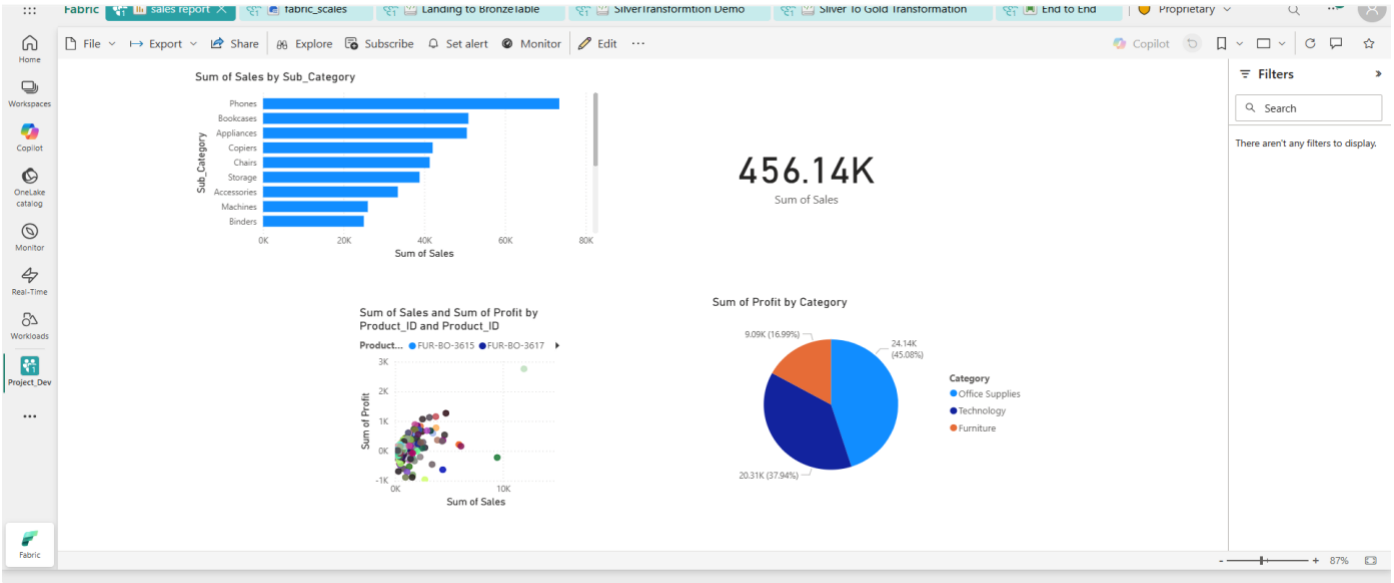
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Sum of Sales by Sub_Category
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9.09K (16.99%)
24.14K (45.08%)
20.31K (37.94%)
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Technology
Furniture
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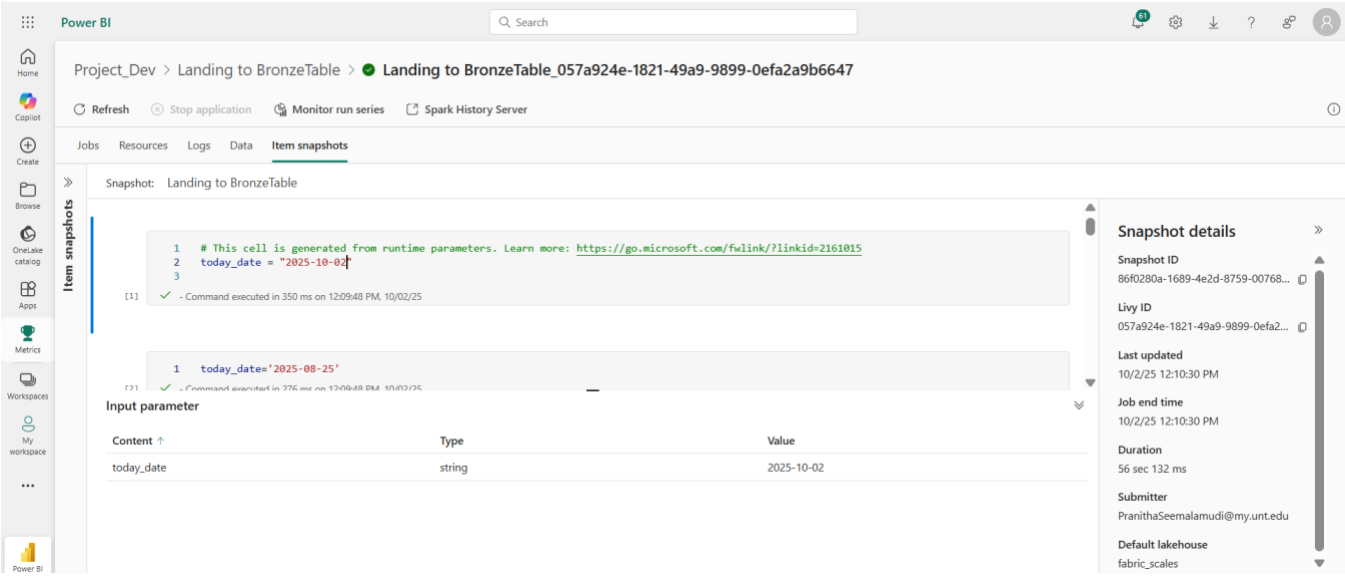
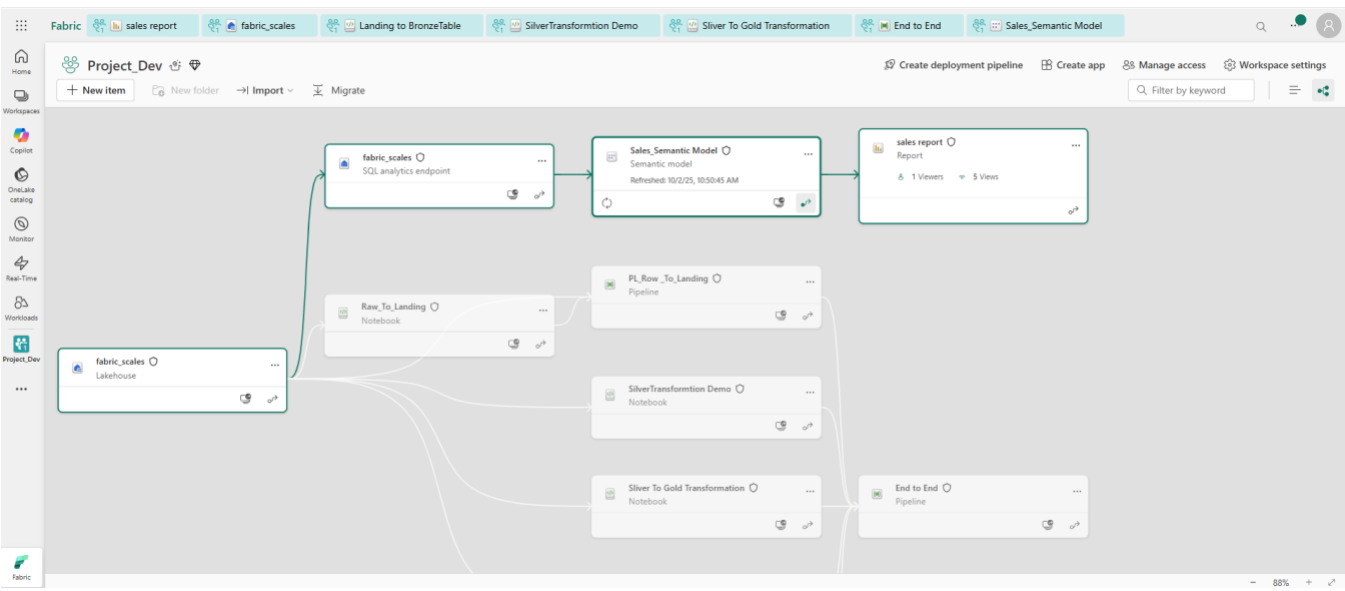
There aren't any filters to display.
87%



Workspace linage:



Note:

This is an automatic model. Updating the data sets and Running the end-to-end pipe lines is enough for analyzing data for todays date. Refresh the sales report for new visualizations.

References and dataset sources:

https://community.tableau.com/s/question/0D54T00000CWeX8SAL/sample-superstore-sales-excelxls

https://www.kaggle.com/code/yorkyong/global-superstore-dataset-data-exploration/input

https://medium.com/the-software-frontier/medallion-architecture-a-comprehensive-overview-f4ef3e8b9974